Deividas Banevicius

Graduate Electrical & Electronic Engineer | Automation · Controls · Embedded Systems

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PERSONAL PROFILE

Electrical and Electronic Engineering graduate (BEng Hons, 2:1) from Manchester Metropolitan University with a strong practical bias: I enjoy taking a system from a blank page to working hardware, then testing and fault-finding it until it performs reliably. My degree and projects span digital design on FPGAs (Verilog), C++ and Python programming, sensor interfacing, PCB layout and oscilloscope-based verification, culminating in a wireless speaker system I designed and built from scratch with sub-100 ms latency. Alongside my studies I work as an automotive diagnostics and electrical technician, where I diagnose real-world electrical, electronic and hydraulic faults on vehicles and industrial machinery. I am now looking for a graduate or junior engineering role in **industrial automation, control systems or embedded engineering**, where I can apply this combination of theory and hands-on skill and continue to learn new technologies such as PLCs and industrial networks.

TECHNICAL SKILLS

Programming	C++, Verilog (HDL), Python, AutoHotkey (AHK) scripting for task automation
Embedded Systems	PIC microcontrollers, ESP32 (Wi-Fi/Bluetooth-enabled MCU); FPGA development on the Intel DE10-Lite
Electronics	Analogue and digital circuit design, logic design, signal analysis, sensor interfacing, hardware testing and troubleshooting, PCB design in KiCad (custom step-down converter)
Control & Power	Control systems theory, three-phase power systems, Fourier transform analysis
Diagnostics	Vehicle electrical diagnostics, ECU/BCM coding (Launch X431, OBDeleven, BimmerCode), wiring loom repair, heavy machinery maintenance (5.5-tonne diesel forklifts)
Software & Tools	Intel Quartus Prime, ModelSim, Multisim, KiCad, VS Code, Git/GitHub, Linux, Windows
Test Equipment	Oscilloscope operation, multimeter, circuit-level and system-level hardware testing

EDUCATION

BEng (Hons) Electrical and Electronic Engineering — 2:1

Oct 2022 – Jul 2025

Manchester Metropolitan University, Manchester

- **Power and Control:** analysed three-phase power systems and modelled the behaviour of feedback control systems, using Fourier transform analysis to move between time and frequency domains — the theoretical foundation for motor drives, process control and automation.
- **Digital Electronics:** designed and analysed digital circuits, and used Multisim to simulate and verify circuit behaviour before building, developing a disciplined design–simulate–test workflow.
- **Final-year individual project** and two FPGA design projects (detailed below) developed my ability to plan, build and document complete engineering systems independently.

A Levels

Sep 2020 – Jul 2022

Beauchamp City Sixth Form, Leicester

Mathematics (B), Physics (C), Computer Science (C). 11 GCSEs at grades 8–6, including Mathematics, English and Computer Science.

ENGINEERING PROJECTS

Wireless Surround Speaker System

Final Year Individual Project

Hardware design · C++ · networked audio · real-time systems

- Designed and built a wireless stereo speaker system from scratch, covering the full system: the electronic hardware and circuitry in each speaker, the software, and the communication link between them.
- Developed C++ software and a server-based architecture so a PC could stream audio to the speakers over a network, handling the flow of data between the host and each speaker unit.
- Achieved **real-time audio transmission with end-to-end latency below 100 ms** — low enough for audio to stay in sync with what is happening on screen — by identifying and reducing delays across the system.
- Managed the project independently from requirements and initial design through to a completed, working system; produced a project plan to schedule the work and a detailed technical report documenting design decisions, testing and results.

FPGA and Digital Electronics Projects

Second and Third Year

Verilog · Intel Quartus Prime · ModelSim · DE10-Lite FPGA board

- Completed two hardware-description projects, writing Verilog, simulating it in ModelSim and then synthesising and implementing it on a DE10-Lite FPGA, including pin assignment and on-hardware troubleshooting.
- Designed an **Arithmetic Processing Unit (APU)** built up from individual logic gates, giving a ground-up understanding of how microprocessors perform arithmetic and logic operations.
- Interfaced multiple sensors — an **ADXL345 accelerometer**, the onboard gyroscope and an **ultrasonic distance sensor** measuring in centimetres — reading their data and converting it into meaningful values shown live on the board's seven-segment hex displays, including alphanumeric characters.
- Generated four switch-selectable waveforms and verified their shape and timing with an oscilloscope, confirming that the simulated design matched real hardware behaviour.

PROFESSIONAL EXPERIENCE

Automotive Diagnostics & Electrical Technician

Family-Owned Import Dealership, Birmingham

- Carry out advanced electrical diagnostics and systematic fault finding on imported Japanese and Hong Kong-market vehicles, using diagnostic scan tools, wiring diagrams and a multimeter to trace faults to their root cause.
- Program and code vehicle control units (ECU/BCM) with Launch X431, OBDeleven and BimmerCode, configuring imported vehicles to work correctly for UK use.
- Perform hardware retrofits, including mileage unit conversions (km to miles), aftermarket Android head unit installations and wiring harness repairs, integrating new electronics safely into existing vehicle systems.
- Maintain and repair mechanical, electrical and hydraulic systems on industrial equipment — for example, diagnosing starter motor and solenoid faults on a 5.5-tonne Heli diesel forklift and returning it to service.
- Work directly within a small family business, taking ownership of jobs from diagnosis to completion and explaining technical issues clearly to non-technical colleagues and customers.

PERSONAL DEVELOPMENT PROJECTS

- **Step-Down Converter PCB (KiCad):** designed the schematic and routed a custom PCB for a DC–DC step-down (buck) converter, building practical skills in component placement, track routing and board layout for power electronics.
- **Python 2D Raycasting Engine:** built a 2D raycasting application from scratch to strengthen programming logic, geometry and graphical rendering skills.
- **Task Automation (AutoHotkey):** write AHK scripts to automate repetitive computer tasks, reflecting a wider interest in automating processes to save time and reduce errors.

AREAS OF INTEREST & INTERESTS

Engineering interests: industrial automation, PLC and control systems, robotics, machine vision, embedded systems, IoT and industrial system integration.

Personal: team-based competitive video games, which build fast communication, teamwork and problem-solving under pressure; badminton at university; teaching myself new skills such as guitar and solving Rubik's Cubes; and exploring the outdoors and rural areas.

References available on request.